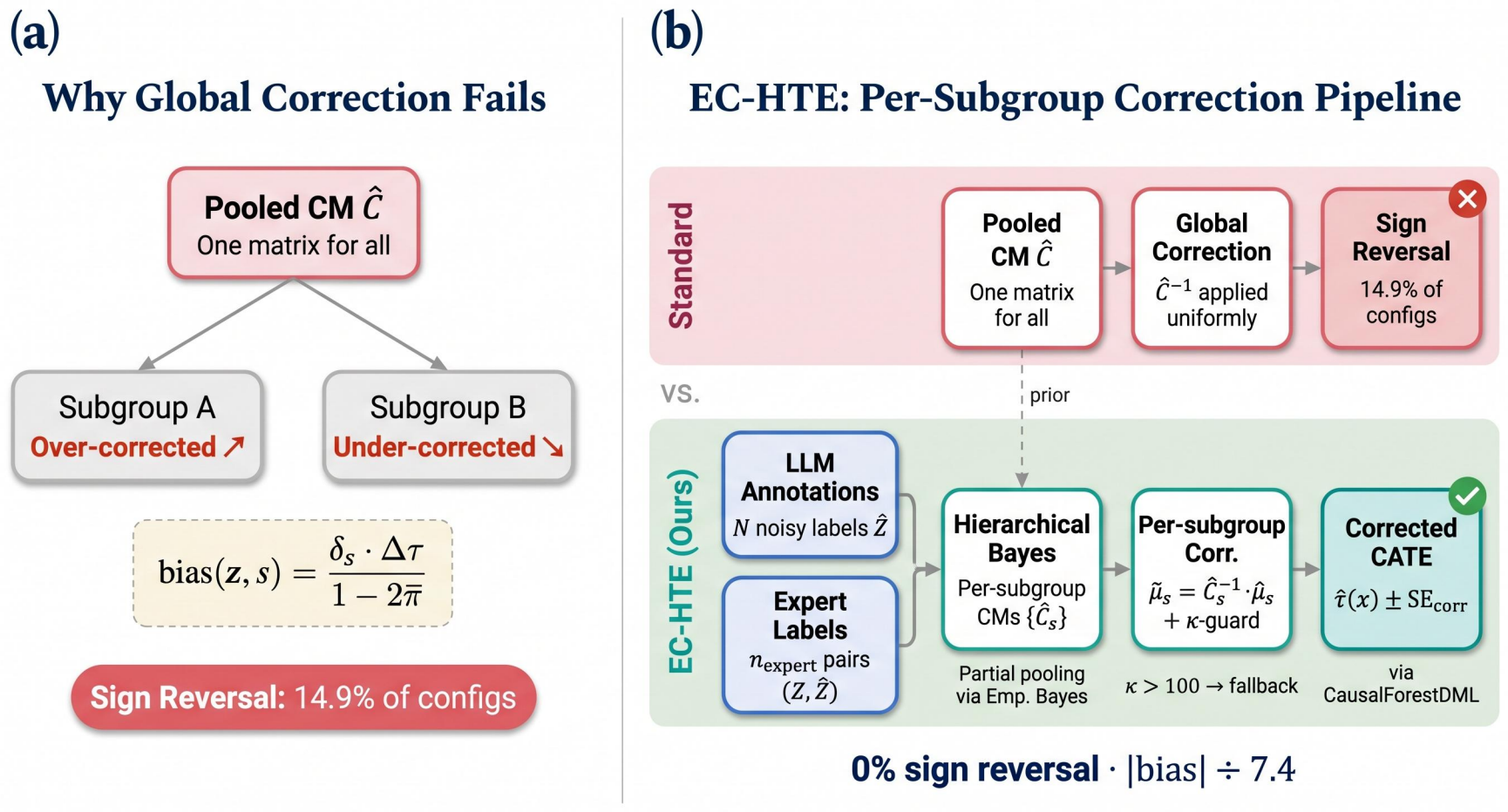

(a)
Why Global Correction Fails
Pooled CM $\hat{C}$
One matrix for all
Subgroup A
Over-corrected ↗
Subgroup B
Under-corrected ↘
$\text{bias}(z, s) = \frac{\delta_s \cdot \Delta\tau}{1 - 2\bar{\pi}}$
Sign Reversal: 14.9% of configs
(b)
EC-HTE: Per-Subgroup Correction Pipeline
Standard
Pooled CM $\hat{C}$
One matrix for all
Global Correction
$\hat{C}^{-1}$ applied uniformly
Sign Reversal
14.9% of configs
VS.
prior
EC-HTE (Ours)
LLM Annotations
N noisy labels $\hat{Z}$
Expert Labels
n_{expert} pairs $(Z, \hat{Z})$
Hierarchical Bayes
Per-subgroup CMs $\{\hat{C}_s\}$
Partial pooling via Emp. Bayes
Per-subgroup Corr.
$\tilde{\mu}_s = \hat{C}_s^{-1} \cdot \hat{\mu}_s$ + κ-guard
$\kappa > 100$ → fallback
Corrected CATE
$\hat{\tau}(x) \pm \text{SE}_{\text{corr}}$
via CausalForestDML
0% sign reversal · |bias| ÷ 7.4